

Every birding app tells you *where*. This one tells you *when*.

Tell it where you are and which bird you want. It gives back the closest hotspot and the hours worth going, using next week's forecast and what birders have actually recorded at those places.

Log. People record what they saw, plus how long they birded, how far they walked, and whether they listed everything they could identify. Those last three are what other logging apps skip, and they are the ones that matter. Without them you have a pile of sightings and no way to judge what any one sighting was worth. The logs also drive a leaderboard, ranked by how many species you found against what the model expected for your effort. Someone birding an hour a week can beat someone birding forty.

Forecast. Pick a bird, pick a spot. The app lists nearby eBird hotspots ordered by your chances rather than by how far away they are, then scores every hour of the coming week. Darker means better odds. Underneath, it breaks down where the number came from, so you can see that leaving at dawn bought you more than the weather did.

Behind it

A penalised logistic model of detection: the chance that someone birding the way you plan to, there, at that hour, reports the species. One model per species, 69 of them.

It is fitted on complete eBird checklists, meaning lists where the birder said they recorded everything they could identify. That matters more than it sounds. If a species is missing from a complete list, that is a real absence. Without it you can count sightings but you cannot compute a probability.

Predictors are the checklist's own effort, the hour, the season, the site, and weather at that site's grid cell for the hours the outing covered. At prediction time the forecast goes in. The coefficients come from what has already been recorded.

The penalty is what lets all 192 hotspots carry their own effect. A spot visited 89 times keeps its own estimate; one visited four times is pulled toward the county average. Without it, a hotspot where the bird was seen every single visit produces an infinite coefficient, which cost roughly half the species when site effects were first added.

Right now it trains on eBird. As people log their own sightings, that feeds a recalibration layer, and the app can lean more on its own data over time.

Scope

This runs on Allegheny County, Pennsylvania, which is a data problem rather than a modelling one. eBird has no bulk endpoint, so getting the denominator costs two API calls per checklist. That is roughly 4,650 calls for a month of one county, 438,000 for a year of Pennsylvania, and over a million for a month of the whole country. Going wider means asking eBird for the Basic Dataset instead of scraping. I still need to work out how to get access to it.

Data: eBird API 2.0, Cornell Lab of Ornithology; ERA5 reanalysis via Open-Meteo. Non-commercial use.